

BHARTENDRA SINGH

 [Github](#)  [Phone](#)  bhartendra750@gmail.com  [LinkedIn](#)  [LeetCode](#)

Technical Skills

Languages: Java(proficient), C++, C

Systems & Programming: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, System Design, Problem Solving, Agile Methodology

Web Technologies: HTML, CSS, JavaScript, React.js, Redux Toolkit, Node.js, Express.js, REST APIs

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, VS Code, Windows

Projects

EduScale | *Node.js, Express.js, MongoDB, Redis, React.js, System Design*

- Designed a **microservices-based architecture** by decomposing the platform into Auth, Course, Enrollment, Payment, and Notification services, enabling independent scaling and better fault isolation.
- Implemented **Redis caching and session management** with **LRU eviction** for frequently accessed data, reducing API response latency by **45%** and lowering database load.
- Designed **optimized database schemas and indexing strategies** following **database-per-service** principles to ensure data consistency and high read/write throughput.
- Applied **load balancing concepts with consistent hashing** to distribute traffic across stateless Node.js services, ensuring high availability and minimal cache re-distribution during horizontal scaling.

StreamPulse | *Java, Redis, MongoDB, OOP, Design Patterns*

- Designed a **real-time distributed streaming platform** using **microservices architecture** with API Gateway, enabling independent scaling of streaming, analytics, and recommendation services.
- Leveraged **Redis** for **live viewer counts, session management, rate limiting, and Pub/Sub messaging**, significantly reducing database load and improving real-time performance.
- Applied **SOLID principles and OOP concepts** in low-level design, using **Factory, Observer, Singleton, and State patterns** to manage stream lifecycle, notifications, and resource sharing.
- Implemented **load balancing with consistent hashing** to distribute live stream traffic across stateless Node.js instances, ensuring high availability and minimal session re-mapping during scaling.

Shortest Path Finder (Dijkstra's Algorithm Visualizer) | *JavaScript, React.js, Graph Theory, DSA*

- Built an interactive pathfinding visualizer with a dynamic **React.js** UI to simulate real-time traversal of nodes and edges.
- Implemented the core logic of **Dijkstra's Algorithm** in plain **JavaScript**, ensuring accurate and responsive path calculation.
- Handled edge cases like negative weights and disconnected components to ensure robustness.
- Used **STL** and **priority queues** in a console-based C++ version to compare performance and validate algorithm efficiency.
- Practiced core **DSA concepts**: graphs, heaps, and performance optimization through hands-on application.

Achievements

- Completed **Full Stack Web Development Certification** from **Apna College**, gaining hands-on experience in MERN Stack.
- Successfully completed **DSA with Java** course by **Apna College**, strengthening problem-solving skills and understanding of core data structures and algorithms.
- Secured **Rank 103** in a weekly coding contest on **Unstop** among **32,000+** participants, demonstrating strong competitive programming and analytical skills.
- Achieved **Top 50** ranking on **LeetCode**, demonstrating strong proficiency in **data structures, algorithms, and problem-solving**.

Education

Graphic Era Deemed University

Bachelor of Technology (B.Tech) in Computer Science

- CGPA: 8.15/10 (Current)

Aug 2023 – Aug 2027

Dehradun, Uttarakhand

AN International School

Higher Secondary Education (Class X & XII)

- Achieved 84.33% in Class XII
- Achieved 79.17% in Class X

Jul 2020 – Jul 2023

Bijnor, Uttarpradesh